

Name: _____ Date: _____ Class: _____

Instructions: Answer all 20 questions. Show your work where needed. Use diagrams, graphs, and coordinate reasoning carefully.

1. Define a point in geometry.

2. How many points are needed to determine a line?

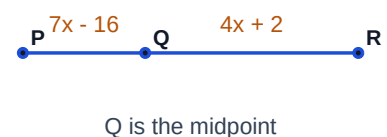
3. How many non-collinear points are needed to determine a plane?

4. What is the intersection of two planes?

5. Name the line through points A and B in two different ways.

6. If M is the midpoint of AB and $AB = 24$, find AM and MB.

7. If Q is the midpoint of PR, and $PQ = 7x - 16$ and $QR = 4x + 2$, find x.



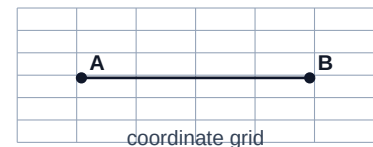
8. Find the midpoint of A(-7, 4) and B(3, -4).

9. If the midpoint of segment XY is (2, -1) and X is (-4, 3), find Y.

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11. Find the distance between $(-2, 5)$ and $(4, 5)$. Use a graph.



12. Find the distance between $A(-4, 1)$ and $B(3, -1)$.

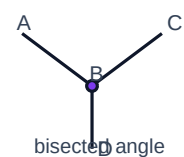
13. Find the midpoint of $G(7, -5)$ and $H(19, -11)$.

14. Find the distance between $E(-7, -2)$ and $F(11, 3)$.

15. On a coordinate plane, a segment has endpoints $(1, 1)$ and $(1, 7)$. What is its length?



16. If ray BD bisects angle ABC and $m\angle ABD = 3x + 5$ while $m\angle DBC = 5x - 11$, find x .



17. If vertical angles are $4x + 12$ and $7x - 24$, find x .

